Paper List

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Abstract

整理一系列文章为了后续能快速的找到内容.

1 World Model

1.1 JEPAs

Training object is (likely) representations, and use CEM to MPC.

- I-JEPA,V-JEPA [1, 2, 3, 4, 5]
- leWM-based [6, 7, 8],
- DINO-WM [9]
- [10], Hierarchical planning

1.2 latent representations

I think that JEPA also helps for building a 'good' representations of model prediction. Here are works that focusing on training the encoder instead of just next-step prediction.

- [11], smoothness for representations
- [12], project DINO token
- *[13], planning in latent*

1.3 WAMs

- [14], predict future
- [15], use optical-flow. *TODO:add lfy's work*
- [16], just denoising w/o predict future
- [17], statistical work to analyze ACTION SPACE
- [18, 19], atom ops of action

1.4 Others

- [20],Point WMs
- [21], dataset for other goals
- [22], testing on Video-gen
- [12], use ODE as training objective
- [23], good pipeline on WM
- [24, 25], ARC-AGI (idk where to put it)
- [26], Probe to find whether WM can find the original laws(e.g.,gravity)

2 Loop

- [27],LoopViT
- [28], use DP to find training-free loops in ViTs
- [29], refine in Diffusion
- [30], LoopDiT
- [31], LoopWM

3 NCA

- [32], Summary
- [33], Diffusion as NCA
- [34, 35], 2D reasoning w/ NCA
- Yitao Xu[1]'s prior work:
  - Growing NCA [36], the original NCA
  - DyNCA [37], dynamic texture synthesis by motion supervision
  - MeshNCA [38], texture synthesis on meshes
  - [39], NCA + appearance decoder, from cells to high-res pixels
  - NoiseNCA [40], noisy seed for spatio-temporal continuity
  - [41], Neural Particle Automata

4 Theory

- [42],question ID/OOD acc using **gradient flow**. Check if u wish to see analyses of subspace
- [43],great work of "capacity" of models[2]
- several works of JEPAs.
  - [7],Gaussian world, how SIGReg work ?
  - [44], LpWM , sprase-space analysis
  - [11], smoothness for representations
  - [45], decode state of WMs to identify
- [46], random matrixs
- [47], Identity in ResNet.Tengyu Ma play

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[1] thx for his wonderful intuition!
[2] here to say, inductive bias is needed for models to learn

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